GEEK ON A LEASH

A source-level Rust talk about physical authority

RUST TUESDAYS • 60 MIN

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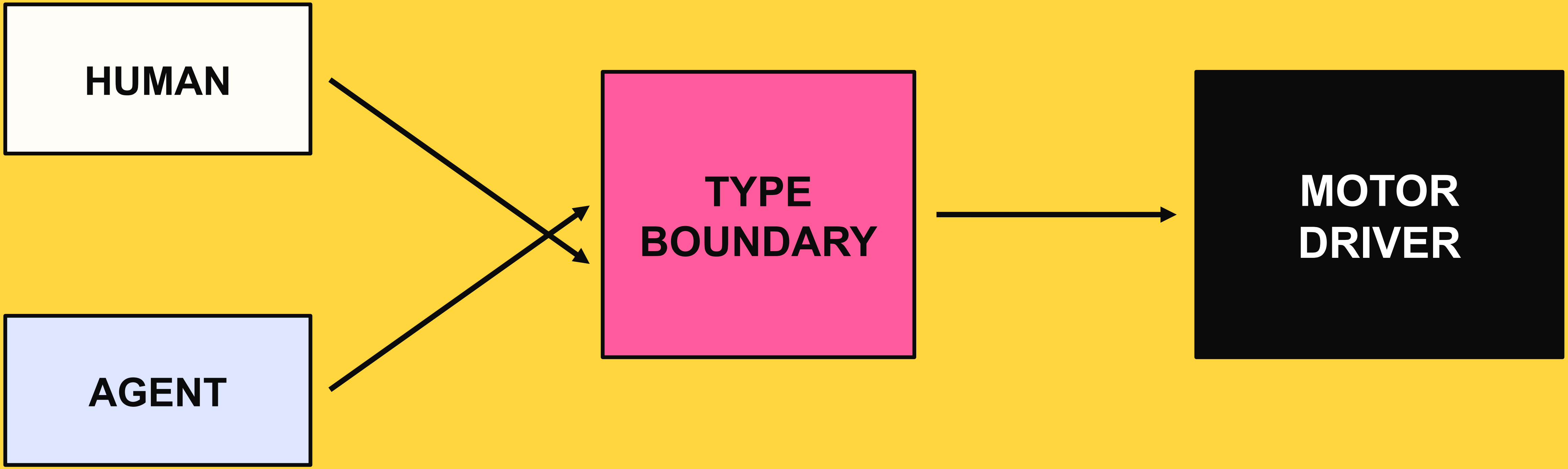
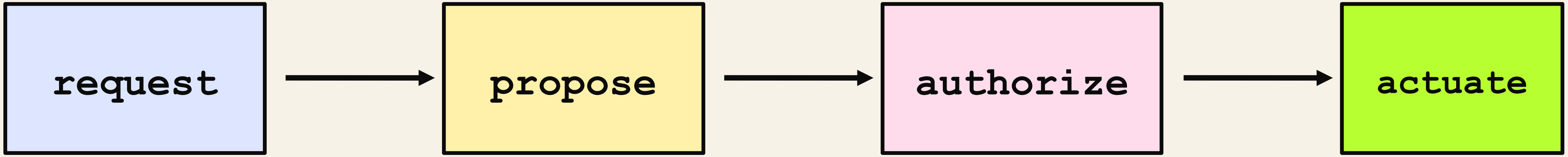
Leash + Pinkie

Who is allowed to write the motors?



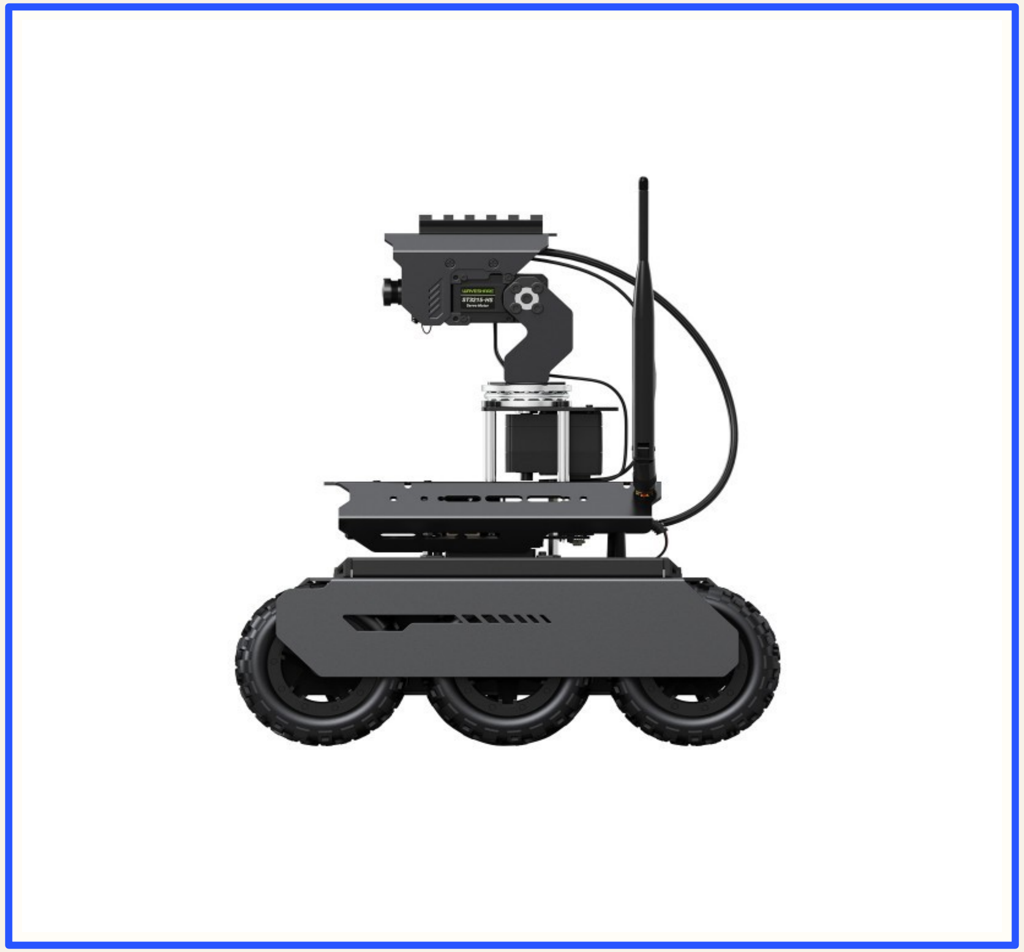
The dangerous bug is ambiguous authority.

The canonical rule is intentionally boring.



No adapter, planner, or GPU kernel may skip a verb.

The types terminate in real hardware.



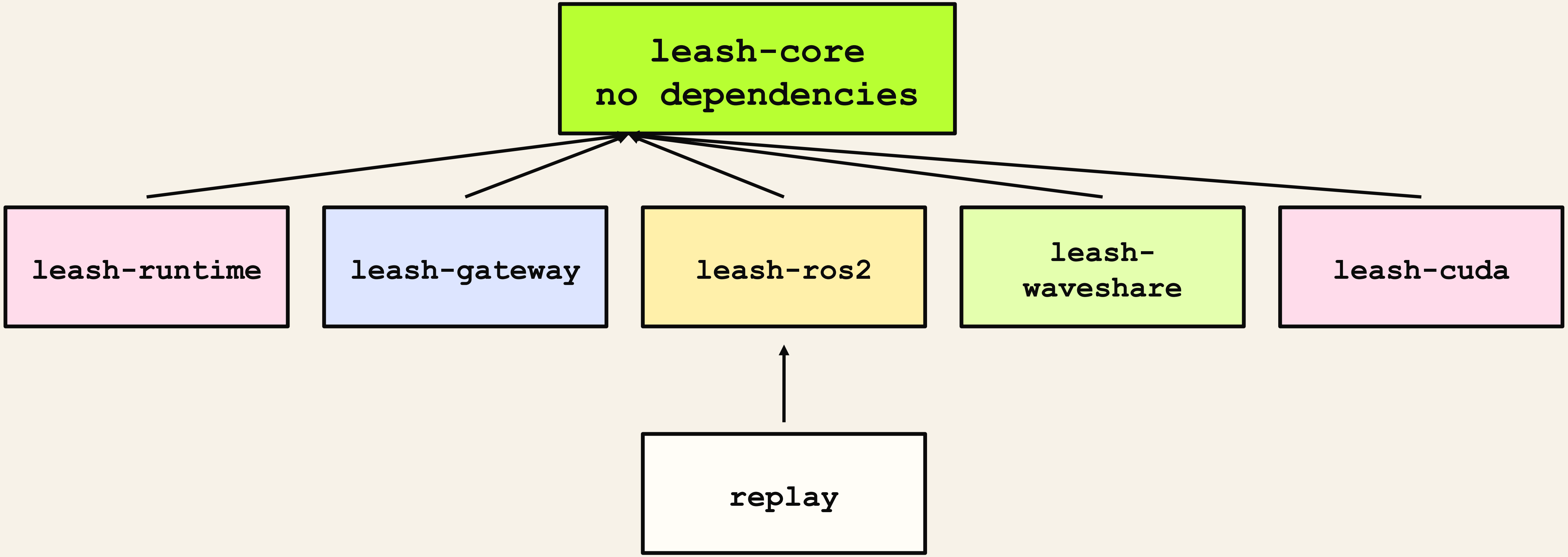
Jetson Orin NX

`serial motor MCU`

LiDAR + camera

A bad abstraction becomes a moving machine.

The crate graph is an authority graph.



Dependencies point inward. Physical authority stays narrow.

Use the compiler to make illegal motion awkward.

NEWTYPE	validate once; keep raw floats out
TYPESTATE	authorization cannot be forged
TRAITS	hardware varies; contract does not
OWNERSHIP	one thread owns physical I/O
DROP	shutdown is part of the type lifecycle

Unsafe is denied by default, then isolated.

crate roots

```
// leash-runtime/src/lib.rs
#![forbid(unsafe_code)]

// leash-waveshare/src/lib.rs
#![forbid(unsafe_code)]

// leash-cuda/src/lib.rs
#![deny(unsafe_code)]

#[cfg(feature = "cuda")]
#[allow(unsafe_code)]
mod device;
```

The default is a compiler error.

The exception is named and feature-gated.

Review the island, not the ocean.

A private field turns validation into a constructor.

leash-core/src/drive.rs

```
#[derive(Debug, Clone, Copy, Default, PartialEq, PartialOrd)]
pub struct NormalizedDrive(f64);

impl NormalizedDrive {
    pub const ZERO: Self = Self(0.0);

    pub fn new(value: f64) -> Result<Self, DomainError> {
        if !value.is_finite() {
            return Err(DomainError::NonFinite("normalized
drive"));
        }
        if !(-1.0..=1.0).contains(&value) {
            return Err(DomainError::OutOfRange("normalized
drive"));
        }
        Ok(Self(value))
    }
}
```

Tuple field is private.

All construction crosses one gate.

Downstream code receives a proof.

Derives are an API decision, not decoration.

leash-core/src/drive.rs

```
#[derive(Debug, Clone, Copy, Default, PartialEq,
PartialOrd)]
pub struct NormalizedDrive(f64);

#[derive(Debug, Clone, Copy, Default, PartialEq)]
pub struct DifferentialDrive {
    pub left: NormalizedDrive,
    pub right: NormalizedDrive,
}

impl DifferentialDrive {
    pub const STOP: Self = Self {
        left: NormalizedDrive::ZERO,
        right: NormalizedDrive::ZERO,
    };
}
```

Copy is safe because the value is tiny and immutable.

PartialEq supports exact protocol tests.

STOP is a canonical, allocation-free value.

One macro stamps the same invariant onto every unit.

leash-core/src/units.rs

```
macro_rules! finite_unit {
    ($name:ident, $label:literal) => {
        #[derive(Clone, Copy, Debug, PartialEq)]
        pub struct $name(f64);

        impl $name {
            pub fn new(value: f64) -> Result<Self,
DomainError> {
                if !value.is_finite() {
                    return
Err(DomainError::NonFinite($label));
                }
                Ok(Self(value))
            }
        }
    };
}
```

macro_rules! removes invariant drift.

Each expansion is still a distinct type.

Meters cannot silently become radians.

NonZeroU64 removes an invalid state from memory.

leash-core/src/time.rs

```
#[derive(Debug, Clone, Copy, PartialEq, Eq, PartialOrd,
Ord, Hash)]
pub struct Sequence(NonZeroU64);

impl Sequence {
    pub fn new(value: u64) -> Result<Self, DomainError> {
        NonZeroU64::new(value)
            .map(Self)
            .ok_or(DomainError::Zero("sequence"))
    }

    pub fn next(self) -> Result<Self, DomainError> {
        self.get()
            .checked_add(1)
            .ok_or(DomainError::Overflow("sequence"))
            .and_then(Self::new)
    }
}
```

Zero cannot be represented.

checked_add makes overflow explicit.

Option says absence is ordinary, not exceptional.

Time arithmetic is checked and domain-specific.

leash-core/src/time.rs

```
#[derive(Debug, Clone, Copy, Default, PartialEq, Eq, PartialOrd,
Ord, Hash)]
pub struct MonotonicNanos(u64);

impl MonotonicNanos {
    pub fn checked_add(self, duration: DurationNanos)
        -> Result<Self, DomainError> {
        self.0
            .checked_add(duration.0)
            .map(Self)
            .ok_or(DomainError::Overflow("monotonic timestamp"))
    }

    pub fn duration_since(self, earlier: Self)
        -> Result<DurationNanos, DomainError> {
        self.0
            .checked_sub(earlier.0)
            .map(DurationNanos)
            .ok_or(DomainError::TimeReversed)
    }
}
```

Monotonic is not wall-clock time.

Subtraction cannot go negative unnoticed.

Overflow is visible in the return type.

Stamped<T>::map moves metadata without cloning it.

leash-core/src/time.rs

```
#[derive(Debug, Clone, PartialEq, Eq)]
pub struct Stamped<T> {
    pub at: MonotonicNanos,
    pub sequence: Sequence,
    pub value: T,
}

impl<T> Stamped<T> {
    pub fn map<U>(self, transform: impl FnOnce(T) -> U) ->
Stamped<U> {
        Stamped {
            at: self.at,
            sequence: self.sequence,
            value: transform(self.value),
        }
    }
}
```

self is consumed: no partial alias survives.

FnOnce permits transforms that consume captures.

Only the payload type changes.

Candidate<C> carries intent, identity, and a deadline.

leash-core/src/drive.rs

```
#[derive(Debug, Clone, PartialEq)]
pub struct Candidate<C> {
    pub id: CommandId,
    pub issued_at: MonotonicNanos,
    pub deadline: MonotonicNanos,
    pub command: C,
}

impl<C> Candidate<C> {
    pub const fn new(
        id: CommandId,
        issued_at: MonotonicNanos,
        deadline: MonotonicNanos,
        command: C,
    ) -> Self {
        Self { id, issued_at, deadline, command }
    }
}
```

C keeps the envelope command-agnostic.

CommandId binds producer epoch and sequence.

A proposal is data, never authority.

Authorization is a value you cannot construct outside core.

leash-core/src/drive.rs

```
#[derive(Debug, Clone, PartialEq)]
pub struct Authorized<C> {
    command_id: CommandId,
    evidence_id: EvidenceId,
    authorized_at: MonotonicNanos,
    command: C,
}

impl<C> Authorized<C> {
    pub const fn command_id(&self) -> CommandId {
        self.command_id
    }

    pub const fn evidence_id(&self) -> EvidenceId {
        self.evidence_id
    }

    pub fn command(&self) -> &C {
        &self.command
    }
}
```

Private fields block struct literals downstream.

Accessors expose facts, not construction power.

The type is a capability token.

The gate consumes a proposal and returns a capability.

leash-core/src/drive.rs

```
pub fn authorize<C>(
    &mut self,
    candidate: Candidate<C>,
    now: MonotonicNanos,
) -> Result<Authorized<C>, SafetyDenial> {
    match self.state {
        SafetyState::Disarmed => return Err(SafetyDenial::Disarmed),
        SafetyState::EStopped => return Err(SafetyDenial::EStopped),
        SafetyState::Faulted => return Err(SafetyDenial::Faulted),
        SafetyState::Ready | SafetyState::Moving => {}
    }

    if candidate.issued_at > now {
        return Err(SafetyDenial::IssuedInFuture);
    }
    if candidate.deadline < now {
        return Err(SafetyDenial::Expired);
    }

    let evidence_id = self.next_evidence_id()?;
    Ok(Authorized {
        command_id: candidate.id,
        evidence_id,
        authorized_at: now,
        command: candidate.command,
    })
}
```

candidate is moved: it cannot be reused accidentally.

match is exhaustive over safety state.

? exits on the first failed proof.

The public API is tested by code that must not compile.

leash-core/src/drive.rs doctest

```
/// \`\`\`compile_fail
/// use leash_core::DifferentialDrive;
/// let _command = DifferentialDrive::new(0.5, 0.5);
/// \`\`\`

/// \`\`\`compile_fail
/// use leash_core::{Authorized, DifferentialDrive};
/// let _forged = Authorized::<DifferentialDrive> {
///     command: DifferentialDrive::STOP,
/// };
/// \`\`\`
```

Documentation becomes a negative contract.

Refactors fail if the forbidden path opens.

The compiler is part of the test harness.

Frame<Tag> stores no tag at runtime—and still separates frames.

leash-core/src/frame.rs

```
pub enum Map {}
pub enum Odom {}
pub enum Base {}
pub enum Sensor {}

#[derive(Debug, Clone, PartialEq, Eq)]
pub struct Frame<Tag> {
    name: FrameName,
    marker: PhantomData<fn() -> Tag>,
}

#[derive(Debug, Clone, PartialEq)]
pub struct Pose2<Tag> {
    pub frame: Frame<Tag>,
    pub x: Meters,
    pub y: Meters,
    pub yaw: Radians,
}
```

Empty enums are type-level labels.

PhantomData ties Tag to ownership/type rules.

No tag bytes are stored in Pose2.

A pose in Odom is not a pose in Map.

leash-core/src/frame.rs doctest

```
/// \`\`\`compile_fail
/// use leash_core::{Frame, FrameName, Map, Meters, Odom,
Pose2, Radians};
/// fn consume_map(_: Pose2<Map>) {}
/// let odom =
Frame::<Odom>::new(FrameName::new("odom").unwrap());
/// let pose = Pose2::new(
///     odom,
///     Meters::new(0.0).unwrap(),
///     Meters::new(0.0).unwrap(),
///     Radians::new(0.0).unwrap(),
/// );
/// consume_map(pose);
/// \`\`\`
```

Same fields do not mean same semantics.

The missing transform becomes a compiler error.

Zero runtime checks; strong compile-time separation.

Supertraits define the minimum hardware capability.

leash-waveshare/src/lib.rs

```
pub trait ControllerIo: Read + Write + Send {}

impl<T> ControllerIo for T
where
    T: Read + Write + Send,
{}

pub trait ControllerIoFactory: Send {
    fn open(&mut self) -> io::Result<Box<dyn ControllerIo>>;
}
```

Read + Write + Send is the complete capability set.

The blanket impl admits every compatible transport.

Box<dyn Trait> erases the concrete serial type.

A closure can be the reconnection factory.

leash-waveshare/src/lib.rs

```
pub trait ControllerIoFactory: Send {
    fn open(&mut self) -> io::Result<Box<dyn ControllerIo>>;
}

impl<F> ControllerIoFactory for F
where
    F: FnMut() -> io::Result<Box<dyn ControllerIo>> + Send,
{
    fn open(&mut self) -> io::Result<Box<dyn ControllerIo>> {
        self()
    }
}
```

FnMut permits reconnect state to evolve.

Send permits ownership transfer to the I/O thread.

The trait object keeps Supervisor non-generic at runtime.

The actuation port chooses its acknowledgement protocol.

leash-runtime/src/supervisor.rs

```
pub trait ActuationAcknowledgement: Send + 'static {
    fn applied(&self) -> bool;
    fn verified_zero(&self) -> bool;

    fn command_id(&self) -> Option<CommandId> { None }
    fn evidence_id(&self) -> Option<EvidenceId> { None }
    fn applied_sequence(&self) -> Option<u64> { None }
}

pub trait ActuationPort: Send + 'static {
    type Acknowledgement: ActuationAcknowledgement;
    type Error: fmt::Display + Send + 'static;

    fn submit_drive(
        &mut self,
        command: Authorized<DifferentialDrive>,
    ) -> Result<(), Self::Error>;

    fn try_acknowledgement(
        &mut self,
    ) -> Result<Option<Self::Acknowledgement>, Self::Error>;
}
```

Associated types bind one Ack and Error to each port.

The port accepts only Authorized<Drive>.

'static permits the whole port to live on a thread.

An equality bound connects the supervisor to the port.

leash-runtime/src/supervisor.rs

```
pub struct CpuSafetySupervisor<A>
{
    handle: SupervisorHandle,
    events: Option<LatestReader<SupervisorEvent<A>>>,
    worker: Option<JoinHandle<()>>,
}

impl<A> CpuSafetySupervisor<A>
where
    A: ActuationAcknowledgement,
{
    pub fn spawn<P>(
        kernel: ControlKernel,
        port: P,
        clock: Box<dyn Clock + Send>,
        config: SupervisorConfig,
    ) -> Result<Self, SupervisorStartError>
    where
        P: ActuationPort<Acknowledgement = A>,
    {
        Self::spawn_inner(kernel, port, clock, config, None)
    }
}
```

A is named once on the supervisor state.

P stays generic only at construction.

Acknowledgement = A is an associated-type equality.

Static dispatch inside; dynamic dispatch at hardware seams.

MONOMORPHIZED

```
fn run_supervisor<P>(
    mut port: P,
    // ...
) where
    P: ActuationPort,
{
    // specialized for P
}
```

ERASED

```
fn controller_loop(
    io: Box<dyn ControllerIo>,
) {
    // one runtime-selected I/O
object
}
```

FAST CONTROL PATH

REPLACEABLE ADAPTER

Choose dispatch based on ownership and change boundaries—not dogma.

move transfers the physical port into one owner thread.

leash-runtime/src/supervisor.rs [abridged]

```
let panic_shared = Arc::clone(&shared);
let thread_shared = Arc::clone(&shared);

let worker = thread::Builder::new()
    .name("leash-cpu-safety".to_string())
    .spawn(move || {
        let _ = thread_shared.worker_thread.set(thread::current());
        let result = std::panic::catch_unwind(
            std::panic::AssertUnwindSafe(|| {
                run_supervisor(
                    kernel, port, clock, config,
                    proposal_receiver, safety_receiver,
                    event_publisher, evidence, shared,
                );
            }),
        );
        if result.is_err() {
            panic_shared.faulted.store(true, Ordering::Release);
            panic_shared.closed.store(true, Ordering::Release);
        }
    })
    .map_err(|error| SupervisorStartError::Thread(error.to_string()))?;
```

move gives the thread exclusive port ownership.

Arc shares state, not mutable hardware access.

catch_unwind converts panic into a fail-closed outcome.

Drop makes thread shutdown part of ownership.

leash-runtime/src/supervisor.rs

```
impl<A> Drop for CpuSafetySupervisor<A>
{
    fn drop(&mut self) {
        self.handle.shared.shutdown.store(true,
Ordering::Release);
        self.handle.shared.wake();

        if let Some(worker) = self.worker.take() {
            let _ = worker.join();
        }
    }
}
```

Dropping sets the shutdown flag.

Wake prevents a sleeping loop from hanging.

Option::take guarantees one join attempt.

Backpressure errors return ownership to the sender.

leash-runtime/src/lane.rs

```rust
pub enum OverflowPolicy {
    RejectNewest,
    DropOldest,
}

pub enum SendOutcome<T> {
    Enqueued,
    ReplacedOldest(T),
}

pub enum SendError<T> {
    Closed(T),
    Full(T),
}

struct LaneState<T> {
    queue: VecDeque<T>,
    high_watermark: usize,
    sent: u64,
    received: u64,
    rejected: u64,
    dropped: u64,
}

struct LaneShared<T> {
    capacity: usize,
    policy: OverflowPolicy,
    closed: AtomicBool,
    state: Mutex<LaneState<T>>,
}
```

The queue has a hard capacity.

Errors preserve the rejected T.

Replacement reports the evicted T.

Overflow policy is an exhaustive match, not a comment.

leash-runtime/src/lane.rs [abridged]

```
pub fn try_send(&self, value: T)
    -> Result<SendOutcome<T>, SendError<T>> {
    if self.shared.closed.load(Ordering::Acquire) {
        return Err(SendError::Closed(value));
    }

    let mut state = lock(&self.shared.state);
    if self.shared.closed.load(Ordering::Acquire) {
        return Err(SendError::Closed(value));
    }

    let outcome = if state.queue.len() == self.shared.capacity {
        match self.shared.policy {
        OverflowPolicy::RejectNewest => {
            state.rejected = state.rejected.saturating_add(1);
            return Err(SendError::Full(value));
        }
        OverflowPolicy::DropOldest => {
            let replaced = state.queue.pop_front()
                .expect("a full bounded queue contains an item");
            state.dropped = state.dropped.saturating_add(1);
            state.queue.push_back(value);
            SendOutcome::ReplacedOldest(replaced)
        }
        }
    } else {
        state.queue.push_back(value);
        SendOutcome::Enqueued
    };
    Ok(outcome)
}
```

Check closed before taking the mutex.

The match documents both overload semantics.

No unbounded queue can consume the device.

Safety requests bypass the ordinary proposal lane.

leash-runtime/src/safety.rs

```
struct SafetyShared {
    stop: AtomicU64,
    estop: AtomicU64,
    closed: AtomicBool,
}

impl SafetySender {
    pub fn request(&self, kind: SafetyKind)
        -> Result<u64, SafetyRequestError> {
        if self.shared.closed.load(Ordering::Acquire) {
            return Err(SafetyRequestError::Closed);
        }
        let sequence = match kind {
            SafetyKind::Stop => increment(&self.shared.stop),
            SafetyKind::EStop => increment(&self.shared.estop),
        }?;
        if self.shared.closed.load(Ordering::Acquire) {
            return Err(SafetyRequestError::Closed);
        }
        Ok(sequence)
    }
}

fn increment(counter: &AtomicU64) -> Result<u64, SafetyRequestError> {
    counter.fetch_update(Ordering::AcqRel, Ordering::Acquire,
        |current| current.checked_add(1))
        .map(|previous| previous + 1)
        .map_err(|_| SafetyRequestError::SequenceExhausted)
}
```

Safety has a dedicated mailbox.

AcqRel makes increment a read-modify-write boundary.

fetch_update prevents sequence wraparound.

The receiver checks E-stop before stop.

leash-runtime/src/safety.rs

```
pub fn try_recv(&mut self)
    -> Result<Option<SafetySignal>, SafetyReceiveError> {
    let estop = self.shared.estop.load(Ordering::Acquire);
    if estop > self.seen_estop {
        let signal = signal(SafetyKind::EStop, self.seen_estop, estop);
        self.seen_estop = estop;
        return Ok(Some(signal));
    }

    let stop = self.shared.stop.load(Ordering::Acquire);
    if stop > self.seen_stop {
        let signal = signal(SafetyKind::Stop, self.seen_stop, stop);
        self.seen_stop = stop;
        return Ok(Some(signal));
    }

    if self.shared.closed.load(Ordering::Acquire) {
        return Err(SafetyReceiveError::Closed);
    }
    Ok(None)
}
```

E-stop wins if both flags are present.

Per-kind counters preserve coalesced requests.

Closed is distinct from no pending signal.

Option::replace makes latest-only semantics explicit.

leash-runtime/src/latest.rs

```
pub fn publish(&self, value: Stamped<T>)
    -> Result<Option<Stamped<T>>, PublishError<T>> {
    let mut state = lock(&self.state);

    if state.last_sequence
        .is_some_and(|sequence| value.sequence <= sequence) {
        state.rejected_out_of_order =
            state.rejected_out_of_order.saturating_add(1);
        return Err(PublishError::SequenceNotIncreasing(value));
    }

    state.last_sequence = Some(value.sequence);
    state.published = state.published.saturating_add(1);
    let replaced = state.value.replace(value);
    if replaced.is_some() {
        state.replaced = state.replaced.saturating_add(1);
    }
    Ok(replaced)
}

pub fn take(&mut self) -> Option<Stamped<T>> {
    let mut state = lock(&self.state);
    let value = state.value.take();
    if value.is_some() {
        state.taken = state.taken.saturating_add(1);
    }
    value
}
```

Stale sequence numbers are rejected.

replace returns the displaced observation.

take moves the newest value out exactly once.

Serde turns the network shape into an enum.

leash-gateway/src/lib.rs

```
#[derive(Debug, Clone, PartialEq, Deserialize, Serialize)]
#[serde(
    tag = "command",
    rename_all = "snake_case",
    deny_unknown_fields
)]
pub enum CommandRequest {
    Authorize {
        operator: String,
        expires_at_ns: u64,
    },
    Drive {
        left: f64,
        right: f64,
        deadline_ns: u64,
    },
    Stop {},
    EStop {},
    ResetEStop { approved: bool },
    SetPlannerActive { active: bool },
    CancelPlanner {},
}
```

Tagged enum gives one explicit command discriminator.

deny_unknown_fields rejects schema drift.

Decode errors enter the typed error path.

map_err and ? preserve context without nesting.

leash-gateway/src/lib.rs

```
pub enum GatewayError {
    InvalidJson(Box<str>),
    InvalidDomain(Box<str>),
    Proposal(SupervisorSubmitError),
    Safety(Box<str>),
    Timeout,
    Supervisor(Box<str>),
    Encode(Box<str>),
}

pub fn decode_and_execute(&self, json: &[u8])
    -> Result<Vec<u8>, GatewayError> {
    let request = serde_json::from_slice(json)
        .map_err(|error| GatewayError::InvalidJson(
            error.to_string().into_boxed_str()
        ))?;
    let response = self.execute(request)?;
    serde_json::to_vec(&response)
        .map_err(|error| GatewayError::Encode(
            error.to_string().into_boxed_str()
        ))
}
```

The enum retains the failure category.

? keeps the happy path visually flat.

Boundary details become owned Box<str> values.

A missing acknowledgement ticket exits immediately.

leash-waveshare/src/lib.rs

```
fn try_acknowledgement(&mut self)
    -> Result<Option<Self::Acknowledgement>, Self::Error> {
    let Some(ticket) = self.pending.front() else {
        return Ok(None);
    };

    match ticket.try_take().map_err(PortError::Submit)? {
        Some(ack) => {
            self.pending.pop_front();
            Ok(Some(ack))
        }
        None => Ok(None),
    }
}
```

let-else handles the precondition at the top.

Empty is normal; disconnected is an error.

front borrows; pop_front happens only after an acknowledgement.

The hardware adapter maps concrete protocol into the trait.

leash-waveshare/src/lib.rs

```
impl ActuationAcknowledgement for CommandAck {
    fn applied(&self) -> bool {
        self.outcome == AckOutcome::Applied
    }
    fn verified_zero(&self) -> bool {
        self.verified_zero
    }
    fn command_id(&self) -> Option<CommandId> {
        Some(self.command_id)
    }
}

impl ActuationPort for WaveshareActuationPort {
    type Acknowledgement = CommandAck;
    type Error = PortError;

    fn submit_drive(
        &mut self,
        command: Authorized<DifferentialDrive>,
    ) -> Result<(), Self::Error> {
        let ticket = self.handle.submit_drive(command)
            .map_err(PortError::Submit)?;
        self.pending.push_back(ticket);
        Ok(())
    }
}
```

Associated types become concrete here.

Authorization remains present at the last adapter.

Hardware errors are not erased prematurely.

Navigation proposals carry their frame in the type.

leash-ros2/src/lib.rs

```
#[derive(Debug, Clone, PartialEq)]
pub struct PathProposal {
    pub frame: Frame<Map>,
    pub at: MonotonicNanos,
    pub poses: Box<[Pose2<Map>]>,
}

#[derive(Debug, Clone, PartialEq)]
pub struct PlanarTransform<Parent, Child> {
    pub parent: Frame<Parent>,
    pub child: Frame<Child>,
    pub at: MonotonicNanos,
    pub x: Meters,
    pub y: Meters,
    pub yaw: Radians,
}

#[derive(Debug, Clone, PartialEq)]
pub struct NavigationGoal {
    pub activity_id: ActivityId,
    pub pose: Pose2<Map>,
    pub received_at: MonotonicNanos,
}
```

Map is encoded in every navigation pose.

Transforms name both parent and child types.

ROS2 is an adapter—not an authority bypass.

Production loads checked kernel bytes; it does not compile at startup.

leash-cuda/src/lib.rs + build.rs [abridged]

```
pub const KERNEL_NAMES: [&str; 7] = [
    "project_occupancy",
    "lidar_transform",
    "spatial_window_transform",
    "collision_sector_reduce",
    "normalize_rgb_u8",
    "predictive_step",
    "predictive_step_metrics",
];

#[cfg(feature = "cuda")]
pub static PREBUILT_FATBIN: &[u8] =
    include_bytes!(concat!(env!("OUT_DIR"),
        "/leash_kernels.fatbin"));

if env::var_os("LEASH_CUDA_REBUILD").as_deref()
    == Some(OsStr::new("1")) {
    rebuild(&source, &output);
} else {
    fs::copy(&prebuilt, &output)
        .unwrap_or_else(|error| panic!(
            "copy prebuilt CUDA artifact: {error}"));
}
```

include_bytes! embeds the exact fatbin in the Rust binary.

SM 8.7 cubin + compute 8.7 PTX are release inputs.

Source and artifact SHA-256 are tested together.

One CUDA thread expands one output cell.

leash-cuda/kernels/leash_kernels.cu

```
extern "C" __global__ void project_occupancy(
    const int8_t* cells,
    int32_t* output,
    uint32_t cell_count,
    uint32_t depth
) {
    const uint32_t index =
        blockIdx.x * blockDim.x + threadIdx.x;
    const uint32_t output_count = cell_count * depth;

    if (index >= output_count) {
        return;
    }

    const int8_t occupancy = cells[index / depth];
    output[index] = occupancy > 0
        ? static_cast<int32_t>(occupancy)
        : 0;
}
```

extern "C" gives Rust a stable, unmangled symbol name.

__global__ means host-called, device-executed.

The bounds mask makes excess launch threads harmless.

Rust validates shape and ownership before the unsafe launch.

leash-cuda/src/device.rs [abridged]

```
let output_count = cells.len()
    .checked_mul(depth as usize)
    .ok_or(ComputeInputError::LengthOverflow)?;
let cell_count = u32::try_from(cells.len())
    .map_err(|_| WorkError::InvalidInput(
        ComputeInputError::LengthOverflow))?;
let launch_count = u32::try_from(output_count)
    .map_err(|_| WorkError::InvalidInput(
        ComputeInputError::LengthOverflow))?;

ensure_capacity(&self.stream,
    &mut self.occupancy_output, output_count)?;

let cells_view = self.occupancy_cells.slice(..cells.len());
let mut output_view =
    self.occupancy_output.slice_mut(..output_count);

unsafe {
    self.stream.launch_builder(&self.project_occupancy)
        .arg(&cells_view)
        .arg(&mut output_view)
        .arg(&cell_count)
        .arg(&depth)
        .launch(LaunchConfig::for_num_elems(launch_count))
}
.map_err(|error| backend_error(
    "launch project_occupancy", error
))?;
```

checked_mul and try_from prevent host-side size corruption.

CudaSlice views bind buffer length and mutability.

unsafe is the claim that Rust arguments match the kernel ABI.

The collision kernel reduces thousands of rays with two atomics.

leash-cuda/kernels/leash_kernels.cu [abridged]

```
const uint32_t index =
    blockIdx.x * blockDim.x + threadIdx.x;
if (index >= count) return;

const float range = ranges_m[index];
if (!isfinite(range)
    || range < range_min_m
    || range > range_max_m) {
    return;
}

const float angle = angle_min_rad
    + static_cast<float>(index) * angle_increment_rad;
const float unwrapped_delta = angle - sector_center_rad;
const float delta = atan2f(
    sinf(unwrapped_delta), cosf(unwrapped_delta));
if (fabsf(delta) > sector_half_width_rad) return;

const float non_negative_range =
    range == 0.0f ? 0.0f : range;
atomicMin(minimum_range_bits,
    __float_as_uint(non_negative_range));
atomicAdd(sample_count, 1U);
```

atan2(sin Δ, cos Δ) wraps angles across ±π.

Non-negative f32 bit ordering makes atomicMin valid here.

The result is advisory; CPU retains final stop authority.

The predictive kernel fuses inference, correction, and learning.

leash-cuda/kernels/leash_kernels.cu [abridged]

```
const uint32_t index =
    blockIdx.x * blockDim.x + threadIdx.x;
if (index >= count) return;

const float previous = state[index];
const float prediction =
    weights[index] * previous + bias[index];
const float bottom_up_error = lower[index] - prediction;
const float top_down_error = previous - top_down[index];

const float next = previous
    + 0.12f * source_precision
        * weights[index] * bottom_up_error
    - 0.05f * top_precision * top_down_error;

state[index] = fminf(4.0f, fmaxf(-4.0f, next));
weights[index] = fminf(1.8f, fmaxf(0.2f,
    weights[index]
        + 0.0005f * bottom_up_error * previous));
bias[index] = fminf(1.0f, fmaxf(-1.0f,
    bias[index] + 0.0001f * bottom_up_error));
```

One thread owns one state, weight, and bias index.

Structure-of-arrays gives adjacent threads adjacent loads.

Fusing three updates avoids intermediate global-memory traffic.

A kernel is only faster when the whole path is faster.

End-to-end p50 on Jetson Orin NX • queue + transfer + launch + sync + readback • each row normalized

WORKLOAD	CPU	CUDA	DECISION
voxel large	1.299 ms	5.896 ms	CPU
lidar large	0.449 ms	0.385 ms	CUDA AFTER SHADOW
spatial combined	1.520 ms	0.534 ms	CUDA AFTER SHADOW
camera large	4.647 ms	3.047 ms	CUDA + GPU CONSUMER
resident cognition	5.353 ms	6.821 ms	CPU

GPU selection is measured policy—not an identity.

CUDA earns compute authority through shadow parity—and can lose it once.

leash-cuda/src/gate.rs [abridged]

```
pub fn execute(&self, job: ComputeJob)
    -> Result<ComputeGateOutcome, WorkError> {
    match self.status().mode {
        GateMode::Cpu => self.execute_cpu(job),
        GateMode::Shadow => self.execute_shadow(job),
        GateMode::Cuda => self.execute_cuda_with_fallback(job),
    }
}

let cpu_started = Instant::now();
let authoritative = execute_before(&self.cpu, job.clone(),
    cpu_started + self.config.deadline)?;
let cpu_elapsed = cpu_started.elapsed();
let cuda_started = Instant::now();
let cuda = execute_before(&self.cuda, job,
    cuda_started + self.config.deadline)?;
let comparison = compare_results(&authoritative, &cuda,
    self.config.tolerance, cpu_elapsed,
    cuda_started.elapsed());

if comparison.matched {
    status.shadow_completed =
        status.shadow_completed.saturating_add(1);
    if status.shadow_completed
        >= self.config.shadow_samples_required {
        status.mode = GateMode::Cuda;
    }
} else {
    status.mode = GateMode::Cpu;
    status.degraded = true;
}
```

48 randomized shadow comparisons; CPU authoritative throughout.

Absolute or relative tolerance: 1e-5; 16 passes per workload.

Mismatch, launch failure, timeout, or panic falls back to CPU.

Determinism and measurements close the loop.

leash-replay/src/lib.rs

```
#[derive(Debug, Clone, Copy)]
struct StableDigest(u64);

impl StableDigest {
    const OFFSET: u64 = 0xcbf29ce484222325;
    const PRIME: u64 = 0x00000100000001b3;

    fn u64(&mut self, value: u64) {
        for byte in value.to_le_bytes() {
            self.0 ^= u64::from(byte);
            self.0 = self.0.wrapping_mul(Self::PRIME);
        }
    }
}

let first = scenario.verify().unwrap();
let second = scenario.verify().unwrap();
assert_eq!(first, second);
```

58.306 µs
p99 transition

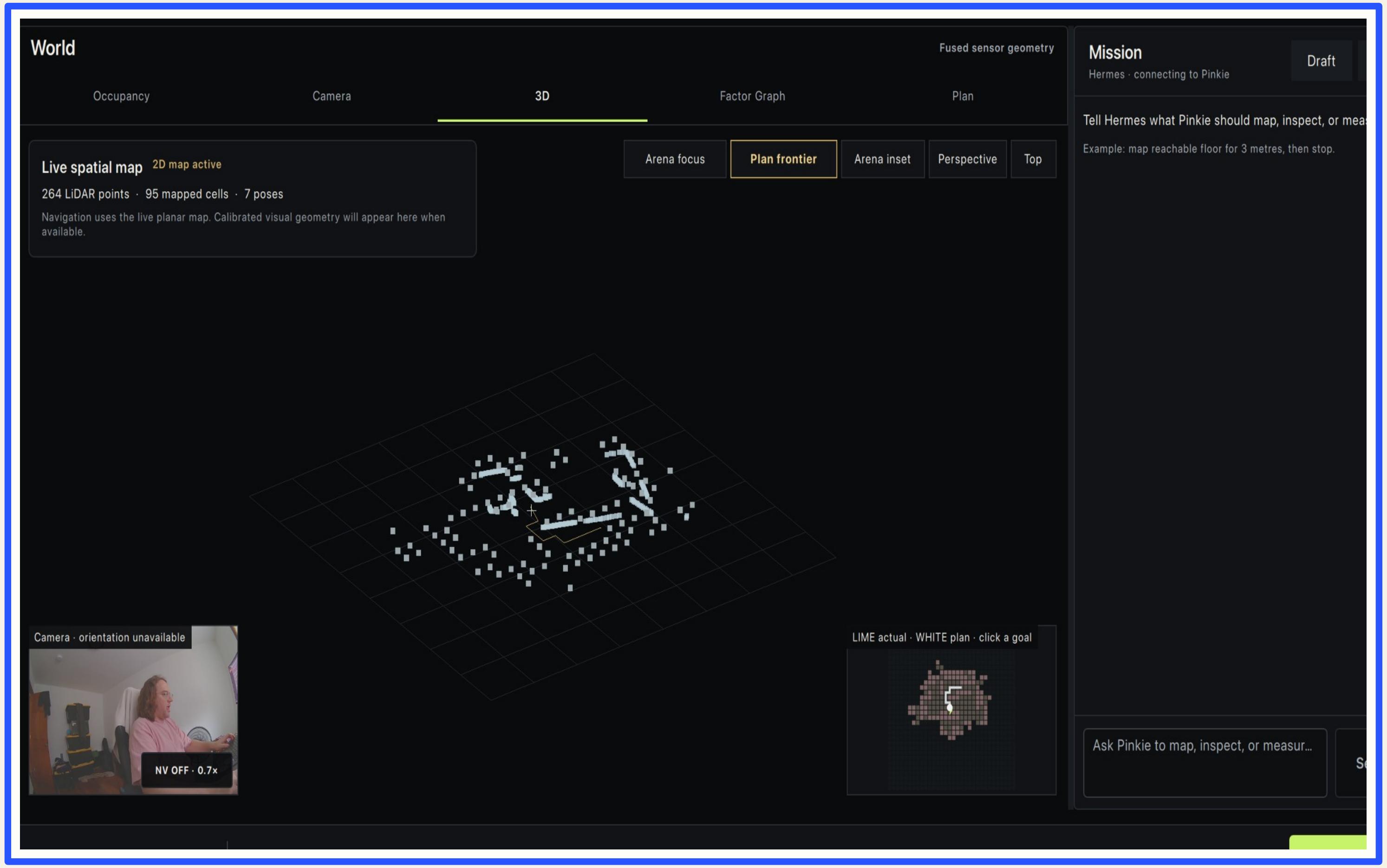
0
deadline misses

110,293
durable records/s

37.622 ms
physical E-stop ack

Same input → same transitions → same digest

Qualia reasons. Leash retains physical authority.



MISSION ONTOLOGY SEMANTICS

↓

PROPOSALS + EVIDENCE

↓

LEASH AUTHORIZES

Demo the boundary, not a heroic autonomy claim.

1. read-only preflight
2. show proposal + evidence
3. approve one ≤ 0.10 pulse
4. verify zero + acknowledgement

operator contract

```
if preflight.is_red() {
    play_recorded_fallback();
    return;
}

assert!(operator_token.is_active());
assert!(pulse.drive <= 0.10);
assert!(pulse.duration <= 500_ms);

let evidence = run_once(pulse)?;
assert_eq!(evidence.final_drive,
STOP);
```

Never improvise movement on stage.

WRITE THE AUTHORITY RULE IN TYPES.

NEWTYPES VALIDATE

TYPESTATE AUTHORIZES

OWNERSHIP ISOLATES

DROP CLOSES

EVIDENCE PROVES

SLIDES + CODE

Direct PDF QR • valid through Sep 8, 2026 12:59 PM ET

github.com/specdog/leash